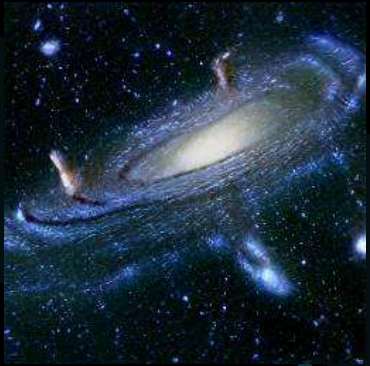
Welcome to the Numerai Universe

A one year journey by QE



Who is QE??

- Dutchie, 44, girlfriend and son, cycling, table tennis
- IT Consultant (cloud / software / data solutions)
- RPI / Arduino robot who likes the cloud and Numerai
- Data Scientist Wannabe

Looking for a challenge…

- Study is nice, but practice is better!

Microsoft Professional Program
Artificial Intelligence Certificate

- Which tournament to pick??

- The most difficult one!!

Start of the journey, 03-2021

Numerai payouts

numerai-cli 0.2

tournament api

Numerati

- Number of Classic Stakes: 2000, 240.000 NMR
- Number of Classic Models : ~6000
- V2 Data: 310 Features, 1 target, 142 eras

numerapi 2.4.5

What is it about??

Overlapping rounds?

Payout factor?

Burn Address?

Neutralization?

Eras?

Voting Ensemble?

NMR staking?

Overfitting?

Correlation?

True Contribution?

MMC?

Nomi target?

20D2L

AUM?

FNC?

TB200?

Stake-Weighted MetaModel?

Corr/TC multipliers?

Era Boosting?

V2/V3/V4?

Stacked Ensemble?

Smart Sortino?

OHwA?

DSC?

CoE?

Training? Validation? Models?

- Curious about NN
- Curious about 'classic' algorithms
- Curious about AutoML
- Try to avoid overfitting
- Try not to lose money!

TensorFlow
Keras
XGBoost
Azure Machine Learning
AutoGluon
AutoKeras
mljar

End Result?

- *Best Corr ranking: 81 (not for long)*
- *Best TC ranking: 2 (not for long)*
- *Earned almost no coins*
- *Learned a Lot!!*

- NN works for a limited time
  - Model Kakkie (V2!)
- AutoML can give quite good results with proper configuration
  - **Models** Villager, xiao_zhishi_2, 3
- Small/medium feature-set is a good start
- Split up that data! -> overfitting
- Try different training data!
  - Validation = Training

Give me some Insights!

tournament api

Classics Dashboard

Compute?

local

Cloud

aws

Azure
ORACLE
Google Cloud Platform
Alibaba Cloud
IBM Cloud

Numerai compute
(~1-xx dollars)

prediction node

kaggle

Hugging Face

Google Colaboratory

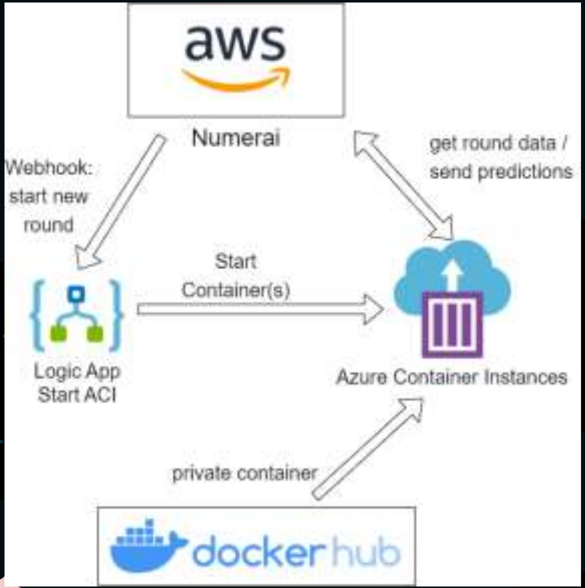
JET BRAINS

gradient by Paperspace

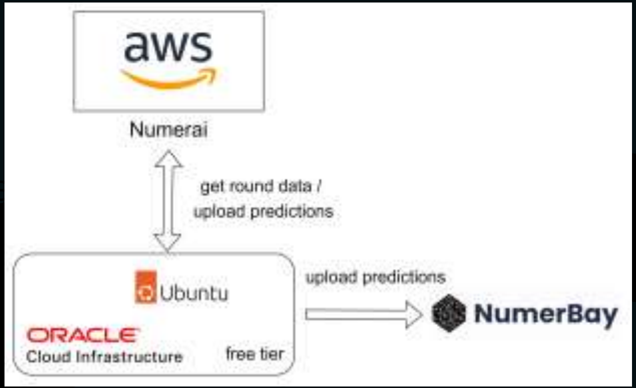
FLOYDHUB

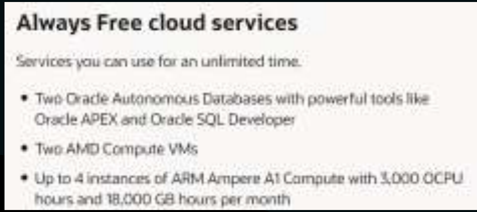
Compute!!

Azure (~1-xx dollars)



Oracle (~0-xx dollars)




Always Free cloud services

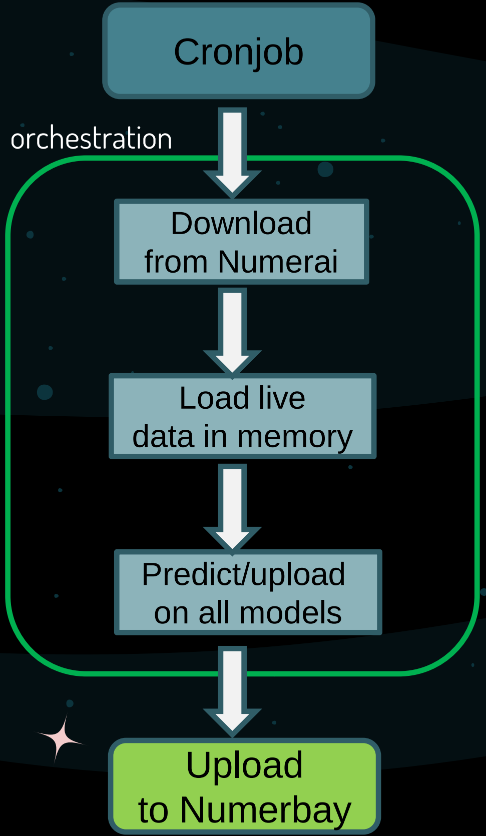
Services you can use for an unlimited time.

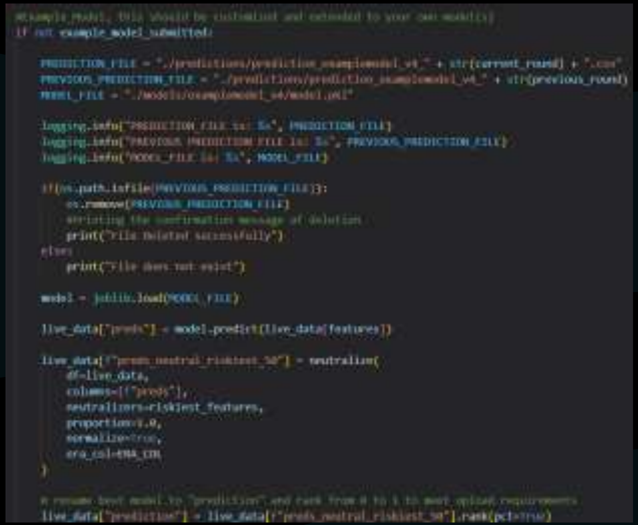
- Two Oracle Autonomous Databases with powerful tools like Oracle APEX and Oracle SQL Developer
- Two AMD Compute VMs
- Up to 4 instances of ARM Ampere A1 Compute with 3,000 OCPU hours and 18,000 GB hours per month

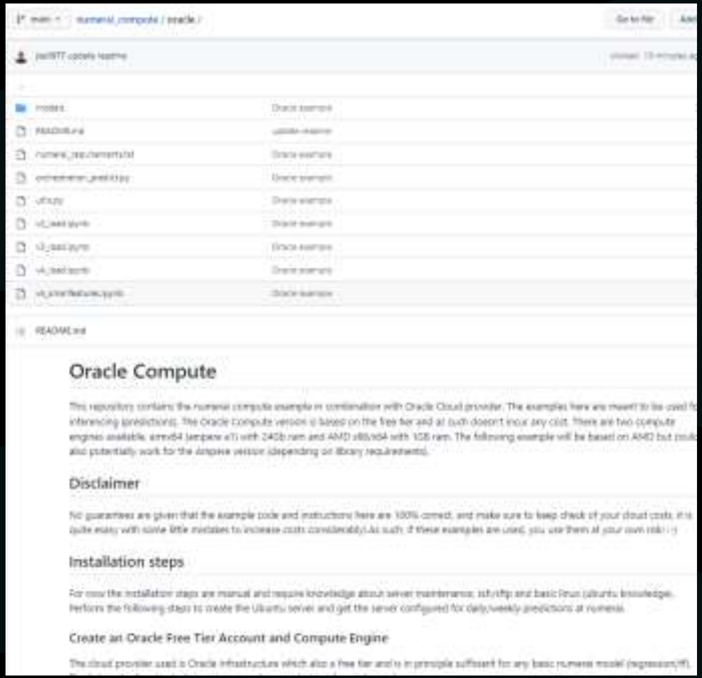
- Own server -> Flexibility
- 24x7 -> Daily Compute
- 2x x86/x64, 1Gb RAM, Swap 12-16Gb
- 1x Arm64 (Ampere A1), 24Gb RAM
- 40 models (TF, regression) < 30 mins

https://github.com/jos1977/numerai_compute

Oracle AMD







https://github.com/jos1977/numerai_compute

Welcome to the Numerai Universe

NumerBay!

Numerai payouts

Awesome Numerai

Numerati

numerai-cli 0.3.4

GraphQL tournament api

numerapi 2.12.1

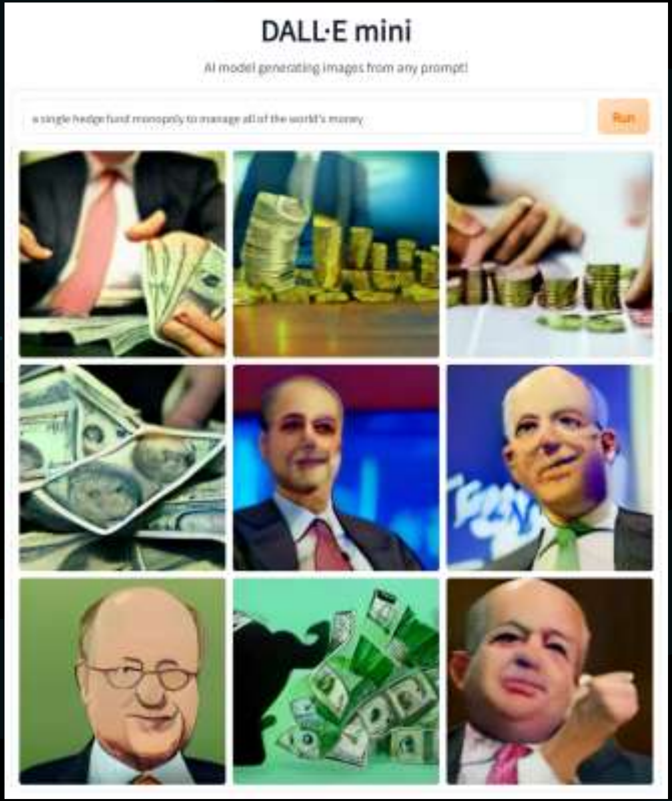
- Number of Classic Stakes: 4400, 700.000 NMR
- Number of Classic Models : ~15000
- V2 Data: 310 Features, 1 target, 142 eras
- V3 Data: 1050 Features, 1 target, 679 eras
- V4 Data: 1194 Features, >10 targets, >679 eras

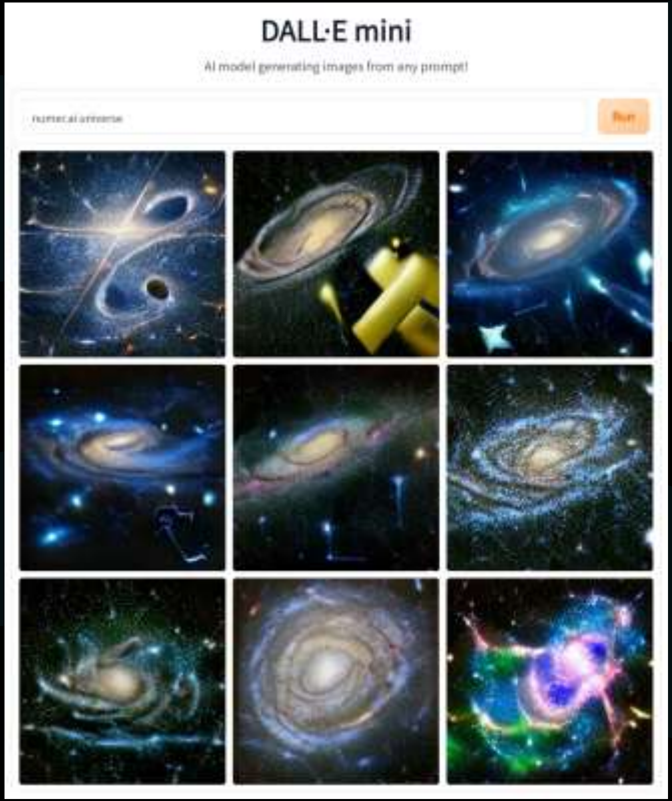
GitHub

numerblox

NUMERAI SIGNALS open signals

Fun with Dall E mini





THANKS!

Any questions?

All the links

Apps / Platforms

- https://numerbay.ai/
- https://github.com/crowdcent/numerblox

API's

- https://github.com/numerai/numerai-cli
- https://github.com/Omni-Analytics-Group/Rnumerai
- https://api-tournament.numer.ai/
- https://github.com/uuazed/numerapi

All the links

Dashboards

- https://dashboard.numeraipayouts.com/
- https://app.powerbi.com/view?r=eyJrIjoiYTI4N2FkY2YtYTliYy00ZjQ5LWE2NjMtMTBjOTkwYTg2NDE2IiwidCI6Ijg3ZDc2ZDQ2LWYxZmYtNDkzMi05MGNiLTUyNzY3Yzg2OTk2ZiIsImMiOjl9
- https://github.com/jos1977/numerai_statistics
- https://www.jofaichow.co.uk/numerati/data.html
- https://www.numerai-insights.com/
- https://share.streamlit.io/yifanxie/numerdash/main/numerdash_app.py
- https://dune.com/aventurine/NumeraiNMR
- https://numerai-earnings.herokuapp.com/

Other

- https://github.com/councilofelders
- https://github.com/peterling7710/NumeraiStarterPack
- https://github.com/jos1977/numerai_compute
- https://gosuto.medium.com/kashi-nmr-borrowing-stake-on-your-model-without-nmr-price-exposure-bc687fcacd9f
- https://gosuto.medium.com/kashi-nmr-lending-earn-apr-without-staking-on-a-model-108c49bc464d
- https://coenumerainewsletter.substack.com/